Atmospheric entry

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

SUCCESS CRITERIA - I can...

1. Compare kinetic energy with a stated heating requirement.
2. Explain the role of surface area relative to mass.
3. Name missing factors in a survival prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ablation” and “thermal energy” to explain your model or result. Sentence starters are optional.

ablation: loss of surface material during heating or erosion

thermal energy: internal energy associated with microscopic motion/interactions

impact: a collision or strike between objects

01 Warm-up: bring back an earlier idea

A process has $\Delta U = -15$ J and $W_{by} = 21$ J.
Find Q and decide whether heat enters or leaves.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

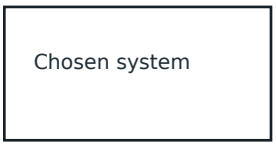
Reference: $K = \frac{1}{2}mv^2$
$Q_{absorbed}$ = fraction absorbed × energy available.
Entry also involves ablation and fragmentation. An energy comparison alone cannot predict survival.

Worked reasoning: A fictional entering object loses 1000 J of mechanical energy; a model assigns 700 J to heated air and 300 J to the object. Those shares are supplied assumptions, not universal atmospheric-entry fractions.

Your turn: Draw the energy budget and identify one measurement needed to test the shares.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Compare a small and a large sphere of the same material. Which has more surface area per unit mass? What does that suggest about surface interactions per kilogram?

Sentence starter: Both … . They differ in … because … .



A B

Marzano check - circle: ① ② ③

04 Apply the physics

An atmospheric-entry simulation starts a hypothetical 10 kg rock at 15 km/s. A supplied heating-and-vaporization requirement is 8.6×10^6 J/kg.
Calculate its kinetic energy, total requirement and their ratio.
Label possible destinations for the energy; explain why the ratio alone cannot predict survival.

Sentence starter: I predict … . My model supports this because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

05 Explain your reasoning

Name at least three additional quantities needed before claiming a particular rock completely ablates or reaches the ground.

Sentence starter: I notice … . This supports … because … .

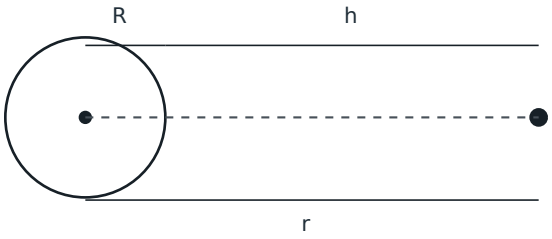
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

For fixed GM, launch radius becomes 16 times as large.
Compare ideal escape speeds and name the equation.
Label the distances or energy levels used in your reasoning.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

If radius doubles at fixed density, derive the area-to-mass multiplier and explain why kinetic-energy-per-mass alone cannot predict a size threshold for survival.
Label energy entering, leaving or stored on the system model.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: I predict … . My model supports this because … .



Radius r Radius 2r

Marzano check - circle: 1 2 3

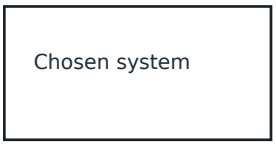
08 STRETCH: test the model

A 2 kg sample has 20 kJ deposited as internal energy. With effective c=500 J/(kg·K), estimate ΔT if no phase change or loss occurs.
State why this is not a survival threshold.
Label energy entering, leaving or stored on the system model.

Word help: estimate = find an approximate value

Sentence starter: Under the assumption …, the evidence … supports … because … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3